Roll Number												

NARULA INSTITUTE OF TECHNOLOGY
An Autonomous Institute under MAKAUT
2025
Odd Semester - 2025-26
PH101 - ENGINEERING PHYSICS

Time: 3 Hrs **Maximum Marks : 60**

Instructions to the candidate:

Figures to the right indicate full marks.
Draw neat sketches and diagram wherever is necessary.
Candidates are required to give their answers in their own words as far as practicable

Part A

Answer any ten from the following, choosing the correct alternative of each question: 10×1=10

1. Given an intrinsic semiconductor doped with donor atoms, determine the approximate position of the Fermi level within the band gap. **(1)** CO2 BL2 PO2

   (a) At donor impurity level. (b) At the middle of the band gap.
   (c) Near conduction band. (d) Near valence band.

2. Analyze the intercepts of a crystal plane that is parallel to the X-axis and determine which Miller indices correctly represent this plane **(1)** CO2 BL2 PO2

   (a) (100) (b) (111)
   (c) (001) (d) (011)

3. An electron and a proton have the same kinetic energy and memp ≈1836. The ratio of their de Broglie wavelengths ($\lambda_e : \lambda_p$) is approximately: **(1)** CO3 BL2 PO2

   (a) 1 : 1 (b) 43 : 1
   (c) 1 : 43 (d) 1836 : 1

4. A material suitable for magnetic storage should have **(1)** CO5 BL2 PO1

   (a) High permeability (b) Low retentivity
   (c) High coercivity and retentivity (d) Low hysteresis loss

5. Which of the following lasers produces red light at 632.8 nm? **(1)** CO1 BL1 PO1

   (a) Ruby laser (b) CO_2 laser
   (c) He-Ne laser (d) Nd:YAG laser

6.	Identify the device in which CRT is most appropriately used (a) OLED display (b) LED display (c) Oscilloscope (CRO) (d) LCD monitor	**(1)**	CO5	BL1	PO1
7.	Compton shift is independent of (a) Wavelength of incident radiation (b) Electron mass (c) Scattering angle (d) Speed of light	**(1)**	CO3	BL2	PO1
8.	Quantum confinement effect is strongest in (a) Quantum dots (b) Thin films (c) Quantum wires (d) Quantum wells	**(1)**	CO5	BL2	PO1
9.	A metal with work function ϕ=3.0 eV is illuminated with light of wavelength 400 nm. The stopping potential is approximately (a) 0.1 eV (b) 0.5 eV (c) 1.0 eV (d) 0.3 eV	**(1)**	CO3	BL2	PO2
10.	The failure of classical physics to explain blackbody radiation is known as (a) Photoelectric effect (b) Ultraviolet catastrophe (c) Compton effect (d) Planck's radiation law	**(1)**	CO3	BL1	PO1
11.	The phenomenon where nanomaterials exhibit different properties at the nanoscale due to the increased dominance of surface effects is called: (a) Bulk confinement (b) Quantum confinement (c) Macroscopic confinement (d) Superconductivity	**(1)**	CO5	BL2	PO1
12.	The MB statistics is applicable for (a) distinguishable particle (b) indistinguishable particle (c) semi distinguishable particle (d) none of these	**(1)**	CO4	BL2	PO1

Part B
(Answer any four of the following) 5x4=20

13. (a)	Explain the phenomenon of Compton effect.	**(2)**	CO3	BL2	PO2
(b)	Calculate the Compton wave length of electron.	**(3)**	CO3	BL1	PO2

14. (a)	Explain the concept of acceptance angle and its importance in optical fiber design.	**(2)**	CO1	BL1	PO1
(b)	For an optical fiber, the refractive index of the core is 1.45 and that of the cladding is 1.42. Calculate the acceptance angle.	**(3)**	CO1	BL2	PO2
15.	State Bragg's Law for x-ray diffraction. A crystal shows a first-order diffraction peak at an angle of 30° for X-rays of wavelength 1.54 Å. Calculate the interplanar spacing.	**(5)**	CO2	BL4	PO2
16.	Answer the following questions				
(a)	Write down the differences between step index optical fiber and graded index optical fiber.	**(3)**	CO1	BL4	PO1
(b)	A10 mw laser beam passes through an optical fiber of 15 km. at the end of the fiber, the signal is 5 mw. Calculate the loss in dB/km.	**(2)**	CO1	BL1	PO1
17.	(a) Using Heisenberg's uncertainty principle, prove that no electron with separate entity can exist in the nucleus of an atom. 3 (b) Why can't Compton effect be visible with ordinary light but can be observed with X-rays? 2	**(5)**	CO3	BL2	PO2
18. (a)	Write the basic principle of data storage in magnetic storage devices.	**(3)**	CO5	BL2	PO1
(b)	What are the advantages of OLED displays over LEDs?	**(2)**	CO5	BL2	PO1

Part C
(Answer any three of the following) 3x10=30

19. (a)	Define the coordination number. Write the coordination number for atoms in an SC and BCC lattice.	**(5)**	CO2	BL2	PO1
(b)	Calculate Atomic Packing Factor (APF) for a simple cubic structure.How it is related to void space?	**(5)**	CO2	BL3	PO1
20. (a)	Derive the relation between Einstein's A and B Coefficients.	**(5)**	CO1	BL2	PO2
(b)	The acceptance angle of a given optical fiber is 45^0. Calculate the numerical aperture of that optical fiber.	**(5)**	CO1	BL2	PO2

			Marks	CO	BL	PO
21.		Answer the following questions				
	(a)	X-rays of wavelength 0.71 Å are reflected from the (110)-plane of a rock salt crystal of Lattice constant a = 2.82 Å. Calculate thecorresponding glancing angle for second Order reflection.	**(5)**	CO2	BL3	PO1
	(b)	Distinguish between intrinsic and extrinsic semiconductors.	**(2)**	CO2	BL1	PO1
	(c)	Give a short note on LCD.	**(3)**	CO5	BL2	PO1
22.		Answer the following questions				
	(a)	What is holography?	**(2)**	CO1	BL1	PO1
	(b)	Discuss the working principle of an optical fiber.	**(5)**	CO1	BL2	PO1
	(c)	An optical fiber is made of glass core of refractive index $\mu 1 = 1.55$ and the cladding of refractive index $\mu 2 = 1.50$ with a core diameter of 50 μm. Calculate the critical angle of the fiber and the acceptance angle of the optical fiber.	**(3)**	CO1	BL3	PO1
23.	(a)	Explain with clear diagrams cathode ray Tube(CRT).	**(5)**	CO2	BL3	PO2
	(b)	State the advantages of LED over LCD .	**(5)**	CO2	BL3	PO2